Youssef Yasser Elsayed Hamed

Junior AI & Software Engineer

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github.com/itz-youssef Portfolio Kaggle Giza, Egypt 12513 (Open to Relocation & Remote)

Profile

Innovative junior Software and AI Engineer dedicated to architecting robust backend infrastructures and state-of-the-art artificial intelligence systems. Highly skilled in leveraging applied deep learning, advanced computer vision, and autonomous agentic workflows to solve complex computational challenges. Committed to applying rigorous software engineering principles to deploy secure, production-grade intelligent solutions.

Education

Cairo University Cairo, Egypt
Bachelor's in Computer Science and Artificial Intelligence *Sep. 2023 – Sep. 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Introduction to Software Engineering, Signals & Systems, Generative Adversarial Networks (GANs), Supervised & Unsupervised Learning, Reinforcement Learning, Robotics & Autonomous Multiagent Systems, Probability & Statistics

Experience

AI Trainee Huawei ICT Academy Program | *Certificate* Sep. 2026 – Oct. 2026
Huawei ICT Academy-Egypt *Remote, Cairo, Egypt*

- Completed the intensive Huawei HCIA-AI training program, mastering **5+** core modules across the full AI development stack.

Computer Vision Engineer Intern | *Certificate* Jul. 2026 – Aug. 2026
National Telecommunication Institute (NTI) *Hybrid, Cairo, Egypt*

- Developed the **Driver Fatigue & Attention Monitoring System [GitHub]**, detecting drowsiness in real-time at **30+ FPS** using **MediaPipe** and a **MobileNetV2 CNN**.
- Engineered a **Flask/SQLite** dashboard with temporal tracking, reducing false alarm rates from normal blinking by **40%**.

AI Training Evaluator (Freelance) Jun. 2026 – Aug. 2026
Outlier *Remote, Cairo, Egypt*

- Evaluated **500+** conversational AI outputs for **RLHF** pipelines, maintaining a **95%+** QA score across naturalness and utility.

Aitronix Internship On-site, New Cairo, Egypt
Artificial Intelligence Intern | *Certificate* *Jun. 2026 – Aug. 2026*

- Developed production-level computer vision and LLM systems, leveraging **FastAPI** to optimize backend data processing times by **30%**.
- Deployed an **n8n**-powered **WhatsApp** bot processing **150+** daily inquiries, reducing manual response latency by **80%**.

Computer Vision Intern *Apr. 2026 – Jun. 2026*

- Built and optimized real-world CV pipelines, integrating **Vision Science** and **RAG** for enhanced context retrieval.
- Engineered a **Behavioral AI Theft Detection** system (**YOLOv11**, **ByteTrack**, **LSTM**), classifying suspicious interactions with **92%** accuracy.
- Deployed **YOLOv8s** models onto enterprise IP cameras via RTSP, achieving **25+ FPS** spatial monitoring across **4** concurrent streams.

Competitions

Competitive Programmer | *Certificate* July 2024
Egyptian Collegiate Programming Contest (ECPC) *Alexandria, Egypt*

- Engineered optimized algorithmic solutions in **C++** under strict time constraints, applying advanced data structures to tackle complex computational problems.
- Collaborated in a high-pressure team environment to prioritize problem selection, rapidly debug logical errors, and maximize competitive performance.

Projects

YAI: Video Safety Analyzer | **[GitHub] [Live Demo]** | *FastAPI, YOLOv10x, CLIP, WebSockets* May 2026

- **End-to-End CV Pipeline:** Engineered an async **FastAPI** server with **WebSocket** streaming for frame-level result delivery and **zero-downtime** frontend decoupling, powering a **5-stage** inference pipeline.
- **Multi-Model Inference Stack:** Integrated **YOLOv10x** + **CLIP** for open-vocabulary detection, and **AnomalyTransformer** to score behaviors across **6** threat categories (loitering, crowd surge, etc.).

APEX: Adaptive Pose & Action Intelligence | **[GitHub]** | *YOLOv8n-pose, LSTM, SQLite* June 2026

- **Action Recognition:** Developed a real-time multi-modal AI tracking system, leveraging **YOLOv8n-pose** to extract **17-keypoint** skeletons and compute **8** critical joint angles per target at high frame rates.
- **Temporal Sequence Modeling:** Engineered a hybrid pipeline fusing rule-based heuristics with an **LSTM** network to process **30-frame** continuous temporal sequences, accurately classifying complex posture transitions.

Pie: AI-Powered Marketing Kit Generator | **[Live Demo]** | *VLMs, Groq LPU, RAG, FastAPI* August 2026

- **Generative AI & RAG:** Built a full-stack AI platform using **Vision-Language Models (VLMs)** to instantly transform product screenshots into structured marketing kits in under **60 seconds**.
- **High-Speed Inference:** Utilized **Groq LPUs** for ultra-low latency generation, designing a custom **Retrieval-Augmented Generation (RAG)** pipeline to dynamically retrieve contextual data.

Behavioral AI Theft Detection System | **[GitHub]** | *YOLOv11x, SAHI, ResNet18 ReID, LSTM* April 2026

- **Robust Detection & Tracking:** Engineered a real-time pipeline actively monitoring **30** high-value classes using **YOLOv11x** and **SAHI** (**1000×1000** tiling), integrated with **ByteTrack** and **ResNet18** ReID.
- **Behavioral Classification:** Extracted **17** dynamic spatial features (grab motion, joint angles) to train an attention-based PyTorch **LSTM** network, classifying suspicious temporal movements across occlusions.

Smart Agriculture Crop Disease Detection | **[Kaggle]** | *ViT, Attention-CNN, DCGAN, PyTorch* Feb - May 2026

- **Automated Diagnosis & Kaggle Bronze Medal:** Built a robust disease classification system trained on **54,000+** images across **15** categories. Benchmarked **14** architectures, with a custom **CNN** achieving **97.5%** accuracy. Awarded a **Kaggle Bronze Medal** for the comprehensive model evaluation notebook.

Certificates & Awards

Huawei Certified ICT Associate (HCIA-AI V4.0) | *Huawei ICT Academy (ID: ICT20261004014238)* Oct. 2026

Microsoft Applied Skills: Secure AI Solutions, Cloud Security & Entra ID | *Microsoft* Sep. 2026

Google AI Essentials | *Google (ID: 8C77QG2AJQAE)* Sep. 2026

Claude Code in Action (Agentic Workflows & LLMs) | *Anthropic (ID: 3i22h3mb75cs)* June 2026

Getting Started with Deep Learning | *NVIDIA (ID: e4Sr2s2OTXC-nnKTTv2bqQ)* Dec. 2025

Master OpenCV Fundamentals for Real-Time CV | *Coursera – EDUCBA* May 2026

Deep Learning Specialization | *DeepLearning.AI (Andrew Ng) – Coursera* Feb. 2026

Technical Skills

Programming & Core: Python, C++, CUDA, Java, JavaScript, SQL, C#, OOP & Design Patterns

Generative AI & LLMs: Vision-Language Models (VLMs), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), LLM Inference APIs (Groq), Agentic AI workflows, Diffusion Models, Zero-Shot Learning

Deep Computer Vision: Multi-Object Tracking, Pose/State Estimation, Instance & Semantic Segmentation, Monocular Depth, Open-Vocabulary Classification, Small-Object Detection (SAHI), Anomaly Detection

MLOps & Cloud Infrastructure: AWS (SageMaker), Google Cloud (Vertex AI, BigQuery), Azure, Kubernetes (K8s), Docker, CI/CD (GitHub Actions), ngrok, MLflow, TensorRT, ONNX, Edge Inference

Frameworks & Ecosystems: PyTorch, Hugging Face, LangChain, LlamaIndex, TensorFlow/Keras, OpenCV, YOLO (v8-v11-v13-v26), CLIP, ByteTrack, MediaPipe, Pandas, Scikit-Learn, Numpy

Data Engineering & Backend: FastAPI, REST APIs, WebSockets, Vector Databases (ChromaDB, Pinecone), PostgreSQL, Supabase, End-to-End Pipeline Architecture

Languages

Arabic: Native
English: Professional Working Proficiency (B2+ / Duolingo English Test: 130)